

"THE GUARDIAN": A MODULAR AI FOR PROACTIVE HUMAN-LAYER CYBERSECURITY ON SOCIAL MEDIA.

Project ID: 25-26J-041

Project Proposal Report

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B. Sc. (Hons) Degree in Information Technology

Department of Information

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

DECLARATION

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Date: 29/08/2023

The above candidate is carrying out this research for the undergraduate dissertation under my supervision.

Supervisor:

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Date: 29/08/2023

ABSTRACT

The widespread growth of scams and phishing on social media platforms exposes users to financial fraud, identity theft, and psychological manipulation. Existing defenses such as spam filters and antivirus software are primarily reactive and focus on traditional communication channels like email, leaving a gap in protecting users from short, informal, and manipulative content in social media feeds. This research proposes the development of the Scam & Phishing Detection Module within The Guardian, a modular, AI-powered browser extension designed to provide real-time, user-centric protection.

The module adopts a dual-path detection pipeline: (i) a URL analysis path that extracts links, checks them against blacklists, and applies heuristic checks for suspicious domains, and (ii) a language analysis path that preprocesses social media text and uses a fine-tuned DistilBERT classifier to detect scam cues such as urgency, impersonation, and financial pressure. Detected risks are returned in a structured JSON format and communicated to the user via a subtle shield icon and tooltip that explains the threat.

A novel contribution of this work is the introduction of a Safe Browsing feature, enabling flagged links to be opened in either a Safe Preview (read-only summary) or Safe Mode (isolated/incognito session), reducing user exposure while preserving autonomy. The system will be evaluated for detection accuracy ($F1 \geq 0.80$), latency (< 500 ms), and user experience ($> 80\%$ positive feedback in usability surveys). This proactive, educational approach addresses both link-based and text-only scams, contributing a practical, scalable solution to the pressing challenge of social engineering attacks on social media.

Keywords: Cybersecurity, Natural Language Processing, Social Engineering, Scam Detection, Phishing, Safe Browsing

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1. INTRODUCTION

Cybersecurity has increasingly shifted from focusing solely on technical exploits to addressing the human layer, where manipulation, deception, and social engineering are exploited by attackers. Among these threats, online scams and phishing on social media platforms have become one of the most prevalent and damaging forms of attack, leading to financial losses, identity theft, and erosion of digital trust [1].

Unlike traditional email phishing, which has been widely studied, social media scams are shorter, noisier, and embedded within informal conversations, making them harder to detect using conventional filters [2].

Scams on platforms such as Facebook, Instagram, WhatsApp, and Telegram often employ psychological triggers, including false urgency (“Act now or lose access”), impersonation of trusted entities (“This is your bank support team”), and emotional manipulation (“I am stranded abroad, please send money”). These tactics exploit trust and cognitive bias rather than technical vulnerabilities, placing the burden on users to recognize deception in real time [3]. Current security tools, including antivirus software, ad-blockers, and built-in browser protections, primarily defend against malware and unsafe downloads. However, they rarely address human-targeted manipulations, leaving a critical gap in protecting everyday social media users [4].

This research proposes the development of the Scam & Phishing Detection Module as part of The Guardian, a modular, AI-powered browser extension that provides real-time, educational alerts rather than censorship. The module follows a dual-path pipeline: (i) a URL analysis path that detects phishing links through blacklist integration and heuristic checks, and (ii) a language analysis path that leverages a fine-tuned Transformer model to identify scam cues in text-only messages. Importantly, this module also introduces a Safe Browsing feature, allowing flagged

links to be opened in either a Safe Preview (read-only metadata summary) or Safe Mode (isolated session), balancing protection with user autonomy.

By combining technical URL checks with linguistic scam detection, and embedding usable security principles, the Scam & Phishing module addresses a significant gap in social media defense. This introduction sets the stage for exploring the background, research gaps, and methodology that underpin the design and evaluation of the proposed solution.

2. BACKGROUND & LITERATURE SURVEY

This section reviews existing work on scam and phishing detection, with a focus on link-based methods, language-based approaches, the specific challenges of social media text, and the importance of usable security in delivering effective interventions.

2.1 Scam and Phishing as Human-Layer Threats

Social engineering is one of the most effective attack vectors, exploiting human trust and psychology rather than technical vulnerabilities [1]. Scams on social media manifest through fraudulent links, impersonation messages, and emotional manipulation. Studies show that scams rely on cognitive triggers such as urgency, authority, and reward to trick victims into revealing credentials or transferring money [2]. While email phishing has been extensively researched, the shift to social media has increased the scale and reach of such attacks. The informal, fast-paced, and conversational nature of social media content makes scams harder to detect using methods designed for structured communication channels like email [3].

2.2 Automated Detection of Phishing Links

Early anti-phishing research focused heavily on URL and domain analysis. Approaches include:

- Blacklists such as PhishTank and Google Safe Browsing provide real-time databases of known phishing domains [4].

- Heuristic checks that identify suspicious patterns such as homoglyph domains (e.g., paypal1.com vs. paypal.com), missing HTTPS, or abnormal redirect chains [5].
- Machine learning classifiers that analyze lexical and structural features of URLs [6].

These methods are fast and effective for known threats, but they fail against zero-day phishing domains and shortened URLs (e.g., bit.ly) that hide the true destination. As attackers evolve, purely link-based detection is insufficient.

2.3 NLP and Language-Based Scam Detection

To address scams without links, researchers have applied Natural Language Processing (NLP) to detect manipulative or fraudulent messages.

- SMS Spam Detection: Datasets such as the Kaggle SMS Spam Collection have been widely used to train classifiers that distinguish legitimate vs. scam messages [7].
- Phishing Email Corpora: The Nazario phishing dataset includes thousands of phishing emails, enabling supervised learning approaches [8].
- Transformer Models: Recent advances show that lightweight models such as DistilBERT retain most of BERT’s performance while being efficient enough for near real-time detection [9]. These models can capture subtle scam cues such as urgency (“act now”), impersonation (“bank support”), and financial solicitation.

However, most of this research targets emails and SMS, not the short, noisy, emoji-rich language of social media. This domain mismatch represents a critical challenge.

2.4 Usable Security and Educational Nudges

Research in usable security emphasizes that overly aggressive or cryptic warnings lead to “security fatigue”, where users ignore alerts altogether [10]. Subtle, educational cues, often called nudges, are more effective in shaping long-term behavior. Instead of censorship, explanations such as “This message uses false urgency, a common scam tactic” improve user awareness and trust [11]. Current phishing tools rarely integrate these principles, focusing instead on blocking access.

The Scam & Phishing Detection Module differentiates itself by embedding these principles into its design: it detects both link-based and text only scams, communicates risk via non-intrusive tooltips, and uniquely provides a Safe Browsing option to minimize user exposure while reinforcing autonomy.

3. RESEARCH GAP

The review of existing literature and current tools highlights several critical gaps that this research aims to address:

- **Overreliance on URL Based Detection:**

Current phishing defenses rely primarily on blacklists and heuristic URL checks (e.g., HTTPS validation, domain similarity). These approaches are limited to link-based scams and fail when attackers use shortened URLs or pure text-based manipulations [4], [5].

- **Limited Adaptation to Social Media Context:**

Most academic work on phishing detection is based on emails (Nazario corpus) or SMS spam datasets (Kaggle). These domains are relatively structured compared to the short, noisy, and informal text of social media, which includes slang, emojis, and code-switching [7]. There is a lack of tools specifically optimized for this environment.

- **Fragmented Detection Approaches:**

Existing tools generally focus on either malicious links or scam language.

Few integrate a dual-path pipeline that combines the strengths of both approaches into a unified, real-time module [6], [8].

- **Neglect of Usable Security Principles:**

Many security systems adopt a blocking or censorship approach, which frustrates users and leads to “security fatigue.” Few phishing detection tools embed educational nudges that explain the risk in clear, accessible language [10], [11].

- **Absence of Safe Browsing Features:**

No existing scam detection system offers users a safe way to explore flagged content, such as a read-only preview or isolated session. Users are forced into a binary choice: ignore the link or risk clicking it.

Gap Statement

There is a lack of a real-time, user-facing scam and phishing detection solution for social media that:

1. Handles both link-based and text-only scams,
2. Is adapted to the linguistic style of social media,
3. Integrates into a modular browser-extension architecture,
4. Provides educational, non-intrusive warnings, and
5. Introduces a novel Safe Browsing feature to minimize user risk while preserving autonomy.

4. RESEARCH PROBLEM

Based on the identified research gaps, this project addresses the following central problems:

How can a dual-path scam and phishing detection module, combining URL analysis with NLP-based scam language classification, be designed and implemented to deliver real-time, educational alerts on social media while enabling users to safely explore flagged content through a Safe Browsing feature?

Sub-Problems

1. URL Extraction and Analysis

- How can URLs be accurately extracted from short, informal social media text, including shortened or obfuscated links?
- To what extent can blacklist services (e.g., PhishTank, Google Safe Browsing) be effectively combined with heuristic checks to detect zero-day phishing domains?

2. Scam Language Identification

- How can an NLP model be fine-tuned to detect scam-related linguistic cues such as urgency, impersonation, and financial solicitation in noisy, emoji-rich text?
- Which preprocessing strategies (slang normalization, typo correction, emoji handling) improve model accuracy for social media content?

3. Hybrid Integration

- How can URL-based detection (fast but narrow) and NLP-based scam detection (contextual but slower) be integrated into a unified pipeline that meets strict latency requirements (<500 ms)?

4. Safe Browsing Enhancement

- How can “Safe Preview” (read-only summary) and “Safe Mode” (isolated session) be incorporated into the workflow without overwhelming the user?
- What usability metrics can demonstrate that these features reduce risk while preserving autonomy?

5. User-Centric Communication

- How should scam detection results be presented in the browser extension (e.g., shield icon + tooltip) to maximize clarity, trust, and educational value while avoiding “security fatigue”?

5. OBJECTIVES

This research aims to address the identified research problem through the achievement of one main objective, supported by specific, measurable sub-objectives.

5.1 Main Objective

To design, implement, and evaluate a real-time Scam & Phishing Detection Module for *The Guardian* that accurately detects both **link-based** and **text-only scams** on social media, provides **educational, non-intrusive warnings**, and introduces a **Safe Browsing feature** to minimize user exposure while preserving autonomy.

Objective ID	Specific Objective	Expected Outcome / Success Criteria	Timeline
SO1	Develop a URL detection and analysis engine using Regex, blacklist checks (PhishTank, Google Safe Browsing), and heuristic rules.	$\geq 90\%$ detection rate on known phishing domains; < 200 ms response time for URL path.	End of Month 4

SO2	Build and fine-tune an NLP model (DistilBERT) to detect scam-related linguistic cues (urgency, impersonation, financial fraud) in informal social media text.	F1-score ≥ 0.80 on a curated test set of scams vs. benign posts.	End of Month 5
SO3	Integrate URL and NLP analysis into a unified hybrid pipeline , returning a structured JSON response with label, confidence, and explanation.	Average end-to-end latency < 500 ms for posts under 280 characters.	End of Month 6
SO4	Implement a Safe Browsing feature with two modes: (i) Safe Preview (read-only metadata summary of flagged links), (ii) Safe Mode (open in isolated/incognito session).	Functional Safe Preview endpoint; $\geq 80\%$ positive usability feedback from pilot test.	End of Month 7
SO5	Conduct a comprehensive evaluation of the module's performance and usability.	Quantitative metrics (precision, recall, F1); qualitative survey (>80% of users find warnings clear and helpful).	End of Month 8

6. METHODOLOGY

This section outlines the systematic approach used to design and implement the Scam & Phishing Detection Module. It describes the overall Guardian system architecture, the internal workflow of the scam module, data collection and preparation strategies, technical requirements, and evaluation plan.

6.1 Overall System Architecture

The Guardian is implemented as a client–server application. A lightweight browser extension acts as the client, while a FastAPI backend hosts four specialized modules: PII & OpSec, Hate Speech, Misinformation, and Scam & Phishing. The workflow is as follows:

1. Text Extraction – The extension extracts visible text from social media posts or messages as the user browses.
2. Secure Transmission – Extracted text is sent to the backend API via HTTPS.
3. Modular Analysis – The backend dispatches text to each detection module. The Scam & Phishing module runs its dual-path analysis.
4. Aggregated Response – All modules return results as structured JSON to the backend.
5. User Notification – The extension displays a subtle shield icon beside risky text; hovering shows a tooltip with an educational explanation and Safe Browsing options.

This modular architecture ensures efficiency, scalability, and clear separation of responsibilities across components.

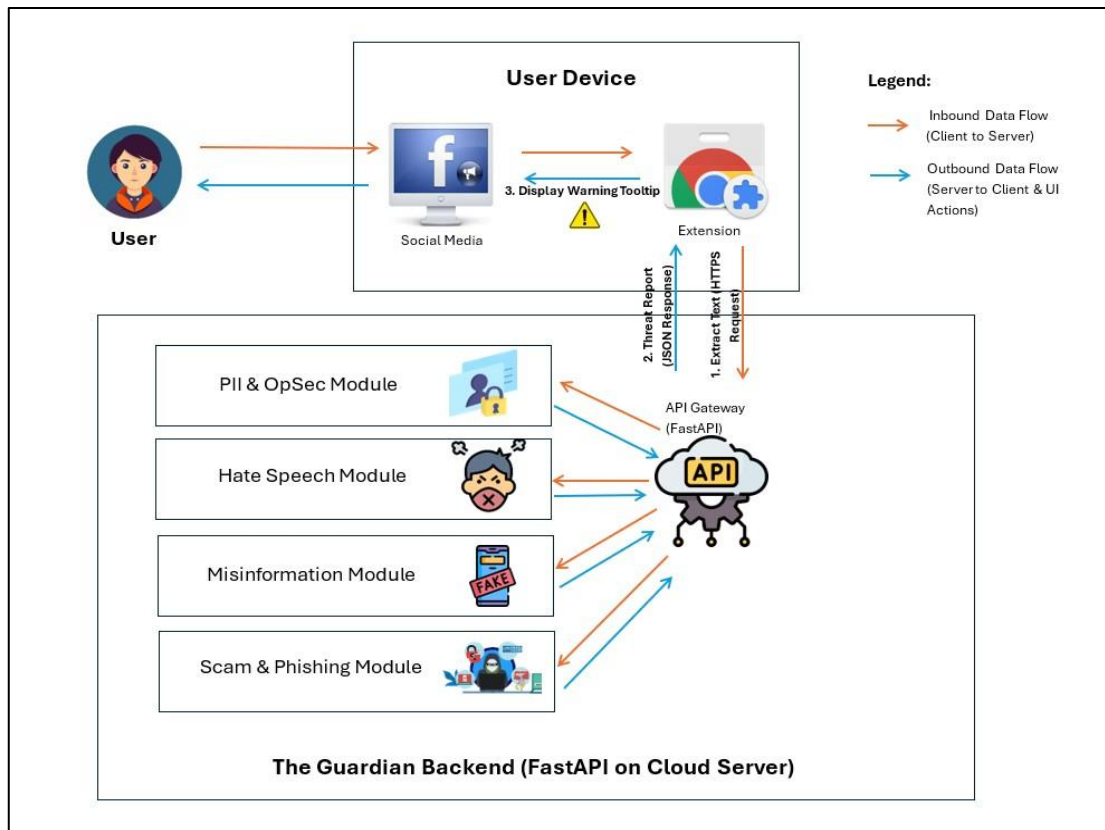


Figure 1: High-level system architecture of "The Guardian," showing the data flow from the client-side browser extension to the backend analysis modules and back.

6.2 Scam & Phishing Module: Design and Implementation

The Scam & Phishing module employs a dual-path pipeline with four stages (Figure 6.2):

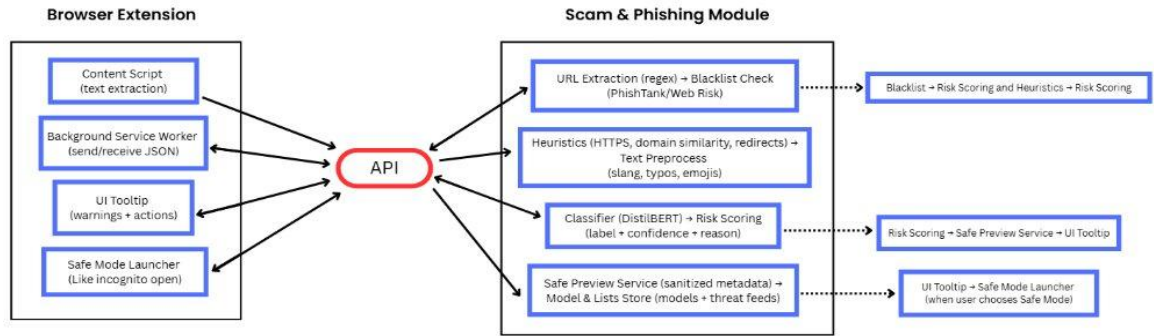


Figure 6.2: Scam/Phishing Module – End to End Flow diagram

- **Stage A: URL Analysis**

- Regex-based URL extraction (including shortened links).
- Blacklist checks via PhishTank and Google Safe Browsing APIs.
- Heuristic checks: HTTPS validation, domain similarity (e.g., paypal.com vs. paypal.com), redirect depth.

- **Stage B: Language-Based Scam Detection**

- Preprocessing: normalize slang, typos, and emojis.
- Classification: fine-tuned DistilBERT model trained on phishing corpora (Nazario, Kaggle SMS spam) + synthetic social-media scam posts.

- Focus on cues such as urgency, impersonation, and financial solicitation.
- **Stage C: Risk Scoring and Unified Output**
 - Merge URL and text outputs into a single JSON response.
 - Output includes label, confidence score, and reason (e.g., “Suspicious URL + urgency pattern”).
- **Stage D: Safe Browsing Feature**
 - Safe Preview: Backend fetches sanitized metadata (final domain, HTTPS status, redirect chain) -> shown to user in a read-only preview.
 - Safe Mode: Option to open flagged links in an incognito/isolated session.

6.3 Data Collection and Preparation

- **Datasets:**
 - Nazario Phishing Corpus (emails with malicious links).
 - Kaggle SMS Spam Dataset (short scam messages).
 - PhishTank blacklist (real-time phishing domains).
- Synthetic Dataset: Create social media–style scam posts with emojis, slang, and mixed-language patterns to reflect real-world noise.

- Partitioning: 70% training, 15% validation, 15% testing.
- Annotation: Posts labeled as scam or benign; categories include impersonation, urgency, financial fraud, and link-based phishing.

6.4 Technical Requirements & Resources

- Programming Language: Python (v3.8+).
- Core Libraries: Hugging Face Transformers, PyTorch, Regex, httpx, FastAPI.
- Environment: Personal laptops (≥ 16 GB RAM) and university labs. GPU (optional, e.g., NVIDIA T4 on Colab) for faster training.
- Version Control: GitHub for collaboration and code management.
- Extension Development: JavaScript, HTML, CSS for browser extension front-end.

6.5 Evaluation Plan

The module will be evaluated using both quantitative and qualitative methods:

- Detection Accuracy: Precision, Recall, and F1-score (target ≥ 0.80).
- Latency: End-to-end pipeline response time < 500 ms for posts under 280 characters.
- Usability Testing: 10+ participants test Safe Preview and Safe Mode; $>80\%$ should rate the tooltips as clear and useful.

- Error Analysis: Manual review of false positives/negatives to identify limitations (e.g., sarcasm, multi-language scams).
- Comparative Benchmarking: DistilBERT vs. baseline methods (e.g., keyword matching).

7. Description of Personnel and Facilities

This section outlines the composition of the project team, their specific roles and responsibilities, and the facilities and resources available to successfully execute the research.

7.1 Personnel

The project team consists of four final-year B.Sc. (Hons) in Information Technology (Cyber Security) undergraduates, each leading a specialized module within *The Guardian*. Roles are distributed to ensure accountability and parallel progress across modules.

M1 – Herath, H. M. G. D. K. (IT22361486)

Project Lead & PII/OpSec Module Lead

- Oversee project coordination, system architecture, and integration.
- Leads development of the PII & OpSec risk detection module, including hybrid NER/Regex model and contextual risk analysis.

M2 – De Alwis, H. P. G. H. S. (IT22317926)

Misinformation Module Lead

- Research linguistic and psychological markers of manipulative content.
- Develops and fine-tunes an AI model for misinformation detection.

M3 – Fernando, M. R. I. (IT22366054)

Hate Speech Module Lead

- Research linguistic patterns of toxic and harmful speech.
- Implements and evaluates the hate speech detection model.

M4 – Vishwa J. R. (IT22916112)

Scam & Phishing Module Lead and UI/UX Lead

- Leads design and implementation of the Scam & Phishing Detection Module.
- Responsible for dual-path detection (URL analysis + NLP-based scam detection).
- Introduces the Safe Browsing feature (Safe Preview and Safe Mode) as a novel contribution.
- Designs the user interface of the browser extension, including wireframes, tooltips, and usability testing.

7.2 Facilities and Resources

The project will be executed using **standard academic facilities** and **open-source resources**, ensuring feasibility without additional cost.

- **Laboratory Facilities:**

Access to SLIIT computing laboratories with adequate networking and hardware infrastructure for development, integration, and testing.

- **Hardware:**

- Personal laptops (≥ 16 GB RAM) for module development.
- University desktops for model fine-tuning and evaluation.

- Optional GPU resources (e.g., Google Colab with NVIDIA T4) for accelerated training.
- **Software:**
 - **Programming Language:** Python (v3.8+).
 - **Core Libraries:** Hugging Face Transformers, PyTorch, spaCy, Regex, FastAPI.
 - **Extension Development:** JavaScript, HTML, CSS for browser extension interface.
 - **Version Control:** Git + GitHub for collaborative development.
- **Data Sources:**

Publicly available datasets (Nazario phishing corpus, Kaggle SMS spam, PhishTank blacklist) supplemented with synthetic social media scam data.

These resources are sufficient to carry out module development, system integration, and evaluation without specialized or high-cost infrastructure.

8. PROJECT REQUIREMENTS

This section defines the requirements of the Scam & Phishing Detection Module, including user expectations, functional and non-functional requirements, use cases, test cases, and initial wireframes.

8.1 User Requirements

The intended users are everyday social media users who are vulnerable to scams and phishing but may lack technical expertise. They require a tool that:

- Operates automatically in the background.
- Detects both malicious links and scam-like text in posts.
- Provides simple, educational warnings rather than censorship.
- Offers a safe way to explore flagged content (Safe Preview and Safe Mode).
- Protects their privacy by not storing personal data.

8.2 Functional Requirements (Fr)

Functional requirements specify the concrete actions the system must be able to perform.

ID	Requirement	Priority
FR1	The system shall extract URLs from user-generated text in real time.	High
FR2	The system shall check extracted URLs against phishing blacklists (PhishTank, Google Safe Browsing).	High

FR3	The system shall apply heuristic checks (HTTPS usage, domain similarity, redirects).	Medium
FR4	The system shall preprocess text (slang, typos, emojis) for NLP analysis.	High
FR5	The system shall classify scam-related text (urgency, impersonation, financial solicitation) using an AI model.	High
FR6	The system shall integrate URL and text results into a unified JSON response.	High
FR7	The system shall display a subtle shield icon next to suspicious posts.	Medium
FR8	The system shall provide an on-hover tooltip explaining the scam risk.	Medium
FR9	The system shall provide a Safe Preview option for flagged links.	High
FR10	The system shall provide a Safe Mode option to open flagged links in an isolated/incognito session.	Medium

8.3 Non-Functional Requirements (Nfr)

Non-functional requirements define the quality attributes and constraints of the system.

ID	Requirement	Metric
NFR1	Performance: Module must respond within 500 ms for posts under 280 characters.	Latency measurement
NFR2	Accuracy: NLP model must achieve $F1 \geq 0.80$ on test dataset.	Model evaluation
NFR3	Privacy: User text must not be stored or logged.	Code review & system audit
NFR4	Usability: Alerts must be minimal and non-intrusive.	Heuristic evaluation, User feedback
NFR5	Security: Extension backend communication must use HTTPS.	Network traffic analysis
NFR6	Modularity: The module must expose a clear API endpoint for integration.	Code structure review

8.4 Use Cases

Use Case 1: Text-only scam detection

- Actor: Social media user.
- Scenario: User reads a post: “Urgent! Reply with your details to claim your prize!”
- System Behavior: Extension extracts text → backend NLP model flags urgency + reward → shield icon appears with tooltip explanation.
- Postcondition: User is warned before responding.

Use Case 2: Phishing link detection

- Actor: Social media user.
- Scenario: Post contains: “Verify your account here: <http://paypal-login.com>”.
- System Behavior: Regex extracts link → blacklist check detects phishing → tooltip shows warning with Safe Preview option.
- Postcondition: User views sanitized preview before deciding.

Use Case 3: Safe Mode browsing

- Actor: Social media user.
- Scenario: User insists on visiting flagged link.
- System Behavior: User clicks Safe Mode → link opens in isolated/incognito session.
- Postcondition: User explores link with reduced risk to main account.

8.5 Test Cases

The following test cases represent a sample of the scenarios that will be used to validate the module's functional requirements.

Test Case ID	Requirement	Test Description	Expected Result
TC-URL-01	FR1, FR2	Input post contains a known phishing URL from PhishTank.	System flags “Urgency Scam” with high confidence.
TC-URL-02	FR3	Input post with domain “paypal.com”.	System flags “Impersonation Scam + Illicit CTA.”
TC-TXT-01	FR4, FR5	Input text: “Urgent! Your account will be closed unless you send details now.”	System flags “Investment Fraud” with explanation.
TC-TXT-02	FR4, FR5	Input text: “Good morning, everyone, have a nice day!”	System returns no scam detected.
TC-JSON-01	FR6	Backend processes input and returns JSON.	JSON includes label, confidence, and reason fields.
TC-UI-01	FR7, FR8	Scam detected.	Shield icon appears; tooltip explains risk.

TC-SAFE-01	FR9	User selects Safe Preview.	Sanitized domain/metadata shown.
TC-SAFE-02	FR10	User selects Safe Mode.	Link opens in isolated/incognito session.

8.6 Wireframes

The user interface for the browser extension alerts is designed to be minimal and non-intrusive. The wireframes in Appendix B illustrate the intended design, showing the appearance of the subtle alert icon and the on-hover tooltip that provides the educational explanation to the user.

9. PROJECT PLAN AND WORK BREAKDOWN STRUCTURE (WBS)

The execution of The Guardian will follow a phased approach, ensuring systematic research, development, integration, and evaluation. Tasks are distributed among team members according to their modules:

- M1: PII & OpSec Module (Lead Integration)
- M2: Hate Speech Module
- M3: Misinformation Module
- M4: Scam/Phishing Module + Browser Extension UI (Me)

A high-level Gantt Chart is provided in Appendix A.

Phase 1: Research And Foundation (Duration: 4 Weeks)

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
1.1	Literature survey on scam and phishing detection approaches (links + text).	M4	Annotated bibliography.	2 Weeks
1.2	Dataset feasibility assessment (Nazario, Kaggle SMS, PhishTank).	M4	Dataset review report.	1 Week
1.3	Finalize scam module architecture and API contract.	M4	Architecture diagram + API specification.	1 Week

Phase 2: Core Module Development (Duration: 6 Weeks Parallel Work)

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
2.1	Implement Regex-based URL extraction.	M4	Python script + test cases.	2 Weeks

2.2	Integrate blacklist checks (PhishTank, Google Safe Browsing API).	M4	Working URL checker.	2 Weeks
2.3	Implement heuristic checks (HTTPS, domain similarity, redirects).	M4	Heuristic engine.	2 Weeks
2.4	Preprocess social media text (slang, typos, emojis).	M4	Preprocessing pipeline.	2 Weeks
2.5	Fine tune DistilBERT classifier for scam detection.	M4	Trained NLP model + logs.	2 Weeks

Phase 3: System Integration and Frontend Development (Duration: 4 Weeks)

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
3.1	Merge URL and text analysis outputs into unified JSON response.	M4	Integrated backend module.	2 Weeks
3.2	Implement Safe Preview feature (sanitized metadata).	M4	FastAPI endpoint /safe_preview.	2 Weeks

3.3	Implement Safe Mode option (incognito/isolated browsing).	M4	Extension code for isolated browsing.	1 Week
3.4	Connect scam module outputs to extension tooltip UI.	M4	End-to-end demo.	1 Week

Phase 4: Testing And Evaluation (Duration: 3 Weeks)

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
4.1	Unit testing of URL pipeline (Regex + blacklist + heuristics).	M4	Test report.	1 Week
4.2	Evaluate NLP classifier (precision, recall, F1-score).	M4	Evaluation report	1 Week
4.3	End to end system testing with backend + extension.	All	Bug reports + performance metrics.	2 Weeks

4.4	User testing of Safe Browsing feature (preview + mode).	M4	User survey results.	1 Week
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Phase 5: Finalization And Reporting (Duration: 4 Weeks)

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
5.1	Draft scam module report section.	M4	Module draft report.	2 Weeks
5.2	Finalize report with integrated module results.	All	Final dissertation report.	1 Week
5.3	Prepare viva presentation (slides + demo).	All	Presentation deck + demo video.	1 Week
5.4	Final code submission + documentation (GitHub).	All	Documented code repository.	1 Week

10. BUDGET AND BUDGET JUSTIFICATION

This project will be completed with no direct financial cost, as it relies on open-source tools, free datasets, and standard computing resources provided by the university and team members.

- **Software Costs:**

All required software is freely available under open-source licenses. This includes Python (v3.8+), FastAPI, Hugging Face Transformers, PyTorch, and Regex libraries. Browser extension development will use HTML, CSS, and JavaScript, which require no licensing fees.

- **Hardware Costs:**

Development and testing will be carried out on personal laptops and university lab computers, both of which meet the requirements ($\geq 16\text{GB}$ RAM). GPU resources such as Google Colab (free tier with NVIDIA T4 GPU) may be optionally used for faster model training.

- **Data Acquisition Costs:**

The datasets required — Nazario phishing corpus, Kaggle SMS spam dataset, and PhishTank blacklist — are all publicly available at no cost. Additional synthetic scam data can be generated programmatically, also incurring no expense.

Therefore, the proposed budget is zero. This ensures the project remains cost-efficient, accessible, and feasible within the scope of an undergraduate research project.

11. COMMERCIALIZATION AND ENTREPRENEURSHIP POTENTIAL

The Guardian is designed with strong commercialization potential as a proactive, educational, and user-focused defense system that addresses a critical gap in the personal cybersecurity market. The Scam & Phishing module directly tackles the most financially damaging human-layer threat: online fraud and phishing.

11.1 Market Opportunity

Scam and phishing attacks represent one of the fastest-growing forms of cybercrime, costing billions annually. By 2027, the number of global social media users is projected to reach nearly six billion, creating a massive attack surface for scammers [14]. Traditional security products, such as antivirus and ad-blockers, focus on technical exploits, leaving human-layer vulnerabilities like social engineering largely unprotected [15].

The Scam & Phishing module fills this gap by detecting both malicious links and scam text, enabling real-time user protection and education. Its Safe Browsing feature further enhances appeal by providing a unique, user-controllable layer of defense absent in mainstream tools.

11.2 Commercialization Pathways

Several commercialization pathways align with industry trends toward subscription-based, user-centric security [16]:

- Freemium Browser Extension (B2C):
 - Free Tier: Core scam/phishing detection (links + text).

- Premium Tier: Subscription (\$2–5/month) unlocking the full suite of Guardian modules, multilingual detection, Safe Browsing options, and customizable alerts.
- Enterprise & Institutional Licensing (B2B):
 - Corporate Security Training: Use as a continuous phishing awareness tool for employees, reducing compliance risks.
 - Educational Integration: Deploy in schools and universities to strengthen digital literacy and scam awareness skills.
- API & Module Licensing (Partnerships):
 - Offer the Scam & Phishing API to cybersecurity vendors, fintech firms, and social media platforms to integrate scam detection into their ecosystems.

11.3 Value Proposition

The Guardian’s commercial strength lies in merging usable security with AI-driven detection [17]:

- Proactive Awareness: Identifies scams before users fall victim.
- Educational Nudges: Provides clear explanations (e.g., “This post uses false urgency, a common scam tactic”) instead of vague alerts.
- Preservation of Autonomy: Allows users to decide, with Safe Preview and Safe Mode as safety nets.
- Scalability: Modular design allows quick addition of new modules as threats evolve.

11.4 Entrepreneurial Impact

The Scam & Phishing module is a high-value entry point for commercialization due to its direct role in preventing fraud and financial losses. By combining technical detection with psychological insight, it strengthens user resilience while providing measurable reductions in scam success rates.

This positions The Guardian as a strong candidate for venture funding, cybersecurity accelerators, and industry partnerships. Serving both consumers and organizations, it bridges security with education offering not only protection but also long-term user empowerment. This dual impact sets it apart from traditional security tools and enhances its entrepreneurial potential.

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Appendix A: Project Gantt Chart

PROJECT TITLE: The Guardian

Group ID: 25-26J-041

Owner: **Lead**
Members: M1 = Group Group/Module A lead - Herath HM G D K (IT22367466),
M2 = Module B lead - De Alwis H P G H S (IT22377926),
M3 = Module C lead - Fernando M P I (IT22386054),
M4 = Module D - UI/UX lead - Vishva J P (IT2239102)

Project Start: Mon, 16/06/2025

Display Week: 1

					16 Jun 2025				23 Jun 2025				30 Jun 2025				7 Jul 2025				14 Jul 2025				21 Jul 2025				28 Jul 2025				4 Aug 2025																		
					16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
TASK	ASSIGNED TO	Milestone	PROGRESS	START	END	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S				
Phase 1: Research and Foundation																																																			
Comprehensive Literature Survey	All	Proposal	100%	11/08/2025	25/08/2025																																														
Research methodology design	M3	Proposal	100%	12/07/2025	17/07/2025																																														
UI/UX wire-frames & mock-ups	M1	Proposal	100%	18/07/2025	22/07/2025																																														
Proposal report writing	All	Proposal Report	100%	16/07/2025	23/08/2025																																														
Proposal Submission (Draft)	M1	Proposal Submission		01/08/2025	29/08/2025																																														
Proposal slide-deck & rehearsal	All	Proposal Presentation		29/08/2025	08/09/2025																																														
Proposal Presentation	All	Proposal Presentation		08/09/2025	12/09/2025																																														
Technology Stack Setup	M1		25%	25/08/2025	31/08/2025																																														
Finalize System Architecture and API 'Contract'	M1		10%	25/08/2025	07/08/2025																																														
Milestone: Project Plan Finalized	M1			07/09/2025	07/09/2025																																														
Phase 2: Core Module Development																																																			
PII/OpSec: Model Development	M1	PP1		08/09/2025	2/11/25																																														
Misinformation: Modle Development	M2	PP1		08/09/2025	2/11/25																																														
Hate Speech: Model Development	M3	PP1		08/09/2025	2/11/25																																														
Scam & Phishing: Model Development	M4	PP1		08/09/2025	02/11/2025																																														
Milestone: All Core Modules Complete	All	Progress Presentatio 1		02/11/2025	02/11/2025																																														
Phase 3: System Integration																																																			
Backend API Development (FastAPI)	M1, M3	PP2		03/11/2025	16/11/2025																																														
Browser Extension Frontend Development	M4, M2	PP2		03/11/2025	16/11/2025																																														
Integrate All Modules with Backend API	M1	PP2		17/11/2025	30/11/2025																																														
Connect Frontend to Backend API	All	PP2		24/11/2025	30/11/2025																																														
Milestone: Integrated System Prototype Ready	All	PP2		30/22/2025	30/11/2025																																														
PP2 deck & integrated demo	All	Progress Presentatio 2																																																	
Phase 4: Testing and Evaluation																																																			
PII/OpSec: Performance Evaluation & Benchmarking	M1	Final		1/12/25	14/12/25																											</																			

Appendix B: Ui Wireframes

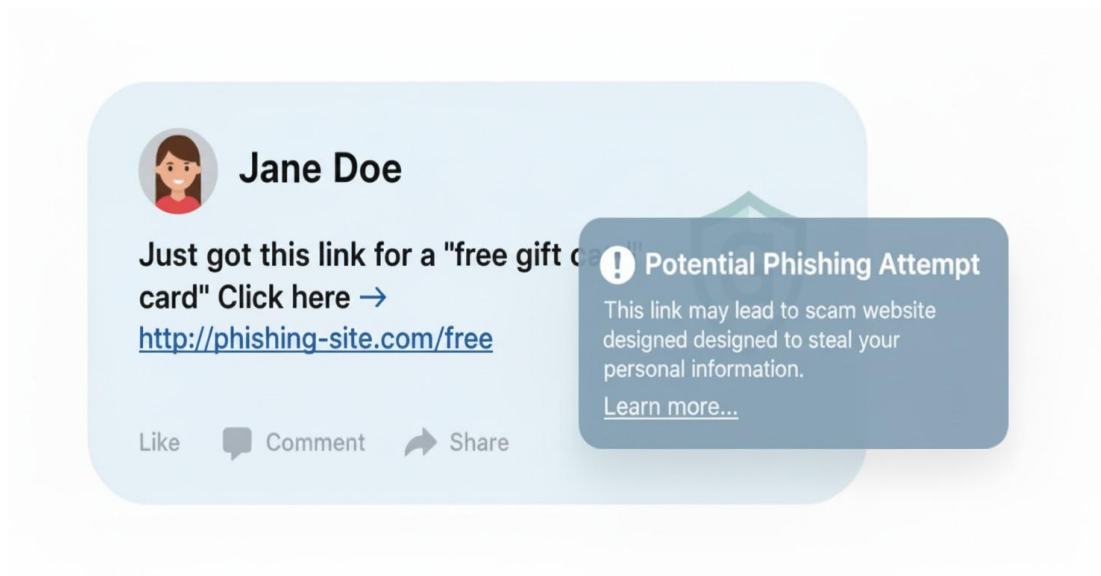
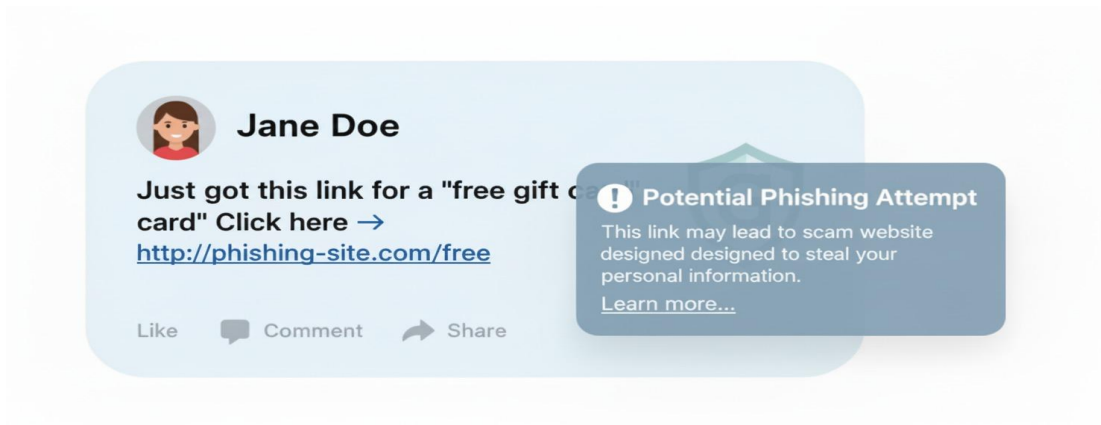


Fig. B.1. **Wireframe of a phishing detection state.** When the system identifies a suspicious link within a social media post, a tooltip appears, providing a concise warning about the potential phishing attempt and offering further explanation, empowering the user to make an informed decision.



Appendix C: Turnitin

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feedback studio Vishwa Rathnayaka IT22916112_RP_Proposal.pdf

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DECLARATION

I declare that this proposal is my own work, and it does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning. To the best of my knowledge, it does not contain any material previously published or written by another person, except where such reference is made in the text.

Name: Vishwa J R
Student ID: IT22916112

Signature: VJR
Date: 29/09/2023

The above candidate is carrying out this research for the undergraduate dissertation under my supervision.

Supervisor:

Signature of Supervisor: [Signature]
Date: 29/09/2023

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